

Manpower woes?

The years when Einstein made his landmark scientific discoveries, he did not even have access to a world class laboratory; he was only working as a minor clerk in the Swiss Patent Office

Manhattan Project

In 1939, the US government began the top secret "Manhattan Project," that aimed at producing a viable atomic bomb



The promise of unlimited Pollution free Power

India's has vast resources of critical elements could be solution for its energy deficit...

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One of the most enchanting ideas for generation of unlimited pollution free power is the one that envisages the usage of antimatter. A design team working on antimatter rocket concepts at Pennsylvania State University has reached the conclusion that if an antimatter rocket were ever built, it would take us to Mars in mere six weeks and we might be back in perhaps four months. Theoretically, a single antimatter power station could meet the energy needs of the entire planet and there would be a little bit to spare for taking care of the requirements of a human colony on Mars. Basically antimatter is the exact opposite of matter, and when the two meet each other, a tremendous amount of energy is released. The idea of antimatter has made a significant impact in the fictional space. No self-respecting space ship in TV or film goes without an antimatter-powered engine. Case in point is USS Enterprise

in the cult movie *Star Trek*. But in the real world, we are yet to overcome the problem of getting hold of the stuff and storing it.

The rise of nuclear power

There is also talk of using nuclear fusion, a process in which two atomic nuclei fuse together to make a heavier metal, and in the process a large amount of energy gets released. Maybe someday human civilization will be able to harness the power of antimatter, and of nuclear fusion, but till that happens, we have to keep utilizing the options we have. Much of the power in the world is generated from coal and petroleum-based plants, but these methodologies lead to the creation of immense amount of greenhouse gases. Hydro-electricity, wind, tidal and solar power are also being explored, but as of now we have been unable to find ways of deriving sufficient energy from such resources. The least polluting and cheapest way of deriving energy is through nuclear fission, during which atoms split into smaller pieces, releasing tremendous burst of energy. At

this point of time, less than 15% of world's electricity is being generated from nuclear power. France is currently generating a maximum amount of its energy needs through nuclear reactors. However, when it comes to nuclear power vessels, US Navy is at the top; it owns majority of the of the world's around 150 nuclear vessels.

The lack of significant development in the nuclear power space must be due to the association of this technology with nuclear weapons. The accidents at Chernobyl and Three Mile Island also contributed in making nuclear technology unacceptable to many sections of society. But if an unbiased and scientific view is taken, then it becomes clear that nuclear energy is as safe as any other way of generating power. The truth is that no lives were lost in the accident at Three Mile Island, and a Chernobyl type accident is physically impossible in a modern reactor. These days we use reactors of self-limiting design, so it is virtually impossible for them to run out of control and explode. Moreover, the prices of natural gas, coal and oil have skyrocketed, so the power generated from such sources is no longer cheap. Society has turned against greenhouse emissions, which result from burning of fossil fuel. If all this is taken into account then nuclear power seems like a much better option; it is cheaper, safer and less polluting.

When Einstein derived his iconic formula $E = MC^2$, he had envisioned a technology capable of producing abundant energy at low cost. So the notion that he was the Father of the Atomic Bomb, which obliterated Hiroshima and Nagasaki in August 1945, is not based on complete truth. If anything, he is the Father of Nuclear Power that meets about 15 per cent of the world's energy needs. The first nuclear reactor in the world was built in 1942, by an American group of researchers, who were being led by the Italian born, scientist Enrico Fermi. Today much of our nuclear power is generated through the use of Uranium. Even though Uranium is a heavy metal, it does not sink to the centre of the earth, and can be found in the continental crust. However, it is never found in natural state; it is mostly found in the form of oxides, such as Uraninite (UO_2) or Pitchblende (U_3O_8). In its natural form, Ura-